Chapter 25
Reset and Boot

25.1 Introduction

This table shows the reset sources that are supported.

Table 25-1. Reset sources

Reset sources	Description
POR reset	• Power-on reset (POR)
System resets	• External pin reset (PIN) • Low voltage detect (LVD) • Watchdog reset • Loss-of-clock (LOC) reset • Loss-of-lock (LOL) reset • Stop mode acknowledge error (SACKERR) • Software reset (SW) • Lockup reset (LOCKUP) • MDM DAP system reset
Debug reset	• JTAG reset

25.2 Reset

This section discusses basic reset mechanisms and sources. Some peripheral modules that cause resets can be configured to cause interrupts instead. See the individual module chapters for more information.

Each reset source has an associated field in the Reset Control Module (RCM) System Reset Status (RCM_SRS) register. Besides immediate system reset, the RCM also supports optional configurable delayed system reset while an interrupt is generated. This provides software an option to perform a graceful shutdown.

The MCU exits reset in RUN mode where the CPU is executing code. See Boot for more details.

25.2.1 Power-on reset (POR)

When power is initially applied to the MCU or when the supply voltage drops below the POR detect voltage (V_{POR}), the POR circuit causes a POR reset condition.

As the supply voltage rises, the LVD circuit holds the MCU in reset until the supply has risen above the LVD low voltage detect threshold (V_{LVD}). Following a POR, the MCU writes 1 to the POR and LVD fields in RCM_SRS register.

Out of POR, both RCM_SRS[POR] and RCM_SRS[LVD] are configured as 1 by MCU.

While normal operation, if device supply goes below VPOR, MCU writes 1 to both RCM_SRS[POR] and RCM_SRS[LVD]. While normal operation, if device supply goes below VLVD (provided LVD is enabled to generate a reset) or VLVR but above VPOR, MCU writes 1 to RCM_SRS[LVD].

25.2.2 System reset sources

Resetting the MCU provides a way to start processing from a known set of initial conditions. System reset begins with the on-chip regulator in full regulation and system clocking generation from an internal reference. When the processor exits reset, it:

- Reads the start stack pointer (SP) (SP_main) from vector-table offset 0
- Reads the start program counter (PC) from vector-table offset 4
- Writes 0xFFFF_FFFF to the Link Register (LR)

The on-chip peripheral modules are disabled and the non-analog I/O pins are initially configured as disabled. The pins with analog functions assigned to them are configured for their analog functions after reset.

During and following a reset, the JTAG pins have their associated input pins configured as:

- TDI as pullup (PU)
- TCK as pulldown (PD)
- TMS as pullup

and associated output pin configured as:

- TDO with no pull-down or pull-up

25.2.2.1 External pin reset

RESET_B is a shared pin (See IO Signal Description Input Multiplexing sheet(s) attached to the Reference Manual). The user must configure the corresponding PORT_PCR register prior to using this pin as RESET_B. On any reset event, the MCU configures the corresponding pad as RESET_B pad. Asserting RESET_B wakes the MCU from any mode. The system clock is required to be available for proper functionality of Reset Pin. During a pin reset, the MCU writes 1 to RCM_SRS[PIN].

25.2.2.1.1 RESET_B pin filter

The RESET_B pin filter supports filtering from both the 128 kHz LPO clock and the bus clock. RCM_RPC[RSTFLTSS], RCM_RPC[RSTFLTSRW], and RCM_RPC[RSTFLTSEL] control this functionality. The filters are asynchronously reset on chip POR. The reset value for each filter assumes the RESET_B pin is negated.

For all stop modes where LPO clock is still active, the only filtering option is the LPO clock filter. The filtering logic either switches to bypass operation or has continued filtering operation depending on the filtering mode selected as described in RCM_RPC register.

The LPO filter has a fixed filter value. The pulses upto 2 LPO128K_CLK cycles will always be filtered. The pulses longer than 3 LPO128K_CLK cycles will always get passed and generate a RESET. The pulses in between 2 to 3 LPO128K_CLK cycles may or may not generate reset based on reset alignment with clock edge.

25.2.2.2 Low voltage detect (LVD)

The chip includes a system for managing low voltage conditions to protect memory contents and control MCU system states during supply voltage variations. The system consists of a POR circuit and an LVD circuit. The LVD system is always enabled in HSRUN and Normal Run. The LVD system is disabled when entering VLPx modes.

The LVD system can be configured to generate a reset upon detection of a low voltage condition by writing 1 to PMC_LVDSC1[LVDRE] . After an LVD reset has occurred, the LVD system holds the MCU in reset until the supply voltage has risen above the low voltage detection threshold. The MCU writes 1 to RCM_SRS[LVD] bit is set following either an LVD reset or POR.

Besides LVD operation, the device also supports LVR(Low Voltage Reset) operation. If the supply voltage falls below the reset trip point (VLVR), a system reset will be generated. LVR system is enabled in all modes. LVDRE has effect on LVD operation only.

Out of POR, both RCM_SRS[POR] and RCM_SRS[LVD] are configured as 1 by MCU.

While normal operation, if device supply goes below VPOR, MCU writes 1 to both RCM_SRS[POR] and RCM_SRS[LVD]. While normal operation, if device supply goes below VLVD (provided LVD is enabled to generate a reset) or VLVR but above VPOR, MCU writes 1 to RCM_SRS[LVD].

25.2.2.3 Watchdog reset

Watchdog timer (WDOG) monitors the operation of the system by expecting periodic communication from the software. This communication is generally known as servicing or refreshing the watchdog. If this periodic refreshing does not occur, the watchdog issues a system reset. The watchdog reset causes the MCU to write 1 to RCM_SRS[WDOG] .

25.2.2.4 Loss-of-clock (LOC) reset

LOC reset is enabled when SOSC clock monitor is enabled (by writing 1 to SOSCCM) and configured to generate a reset on error (by writing 1 to SOSCCMRE). If the error bit (SOSCERR) is written to 1 and LOC reset is enabled (as mentioned above), the MCU resets. The MCU writes 1 to the LOC field to indicate this reset source.

NOTE

To prevent unexpected loss of clock reset events, all clock monitors should be disabled before entering any low power modes, including VLPR.

25.2.2.5 Loss-of-lock (LOL) reset

LOL reset is enabled when PLL clock monitor is enabled (by writing 1 to SCG_SPLLCSR[SPLLCM]) and configured to generate a reset on error (by writing 1 to SCG_SPLLCSR[SPLLCMRE]). If the error bit (SCG_SPLLCSR[SPLLERR]) is written to 1 and LOL reset is enabled (as mentioned above), the MCU resets. The MCU writes 1 to RCM_SRS[LOL] field is set to indicate this reset source.

NOTE

This reset source does not cause a reset if the chip is in any Stop mode.

SPLL also flags LOL in case if the reference to SPLL goes faulty. So while using SPLL, any failure in reference clock source, i.e., SOSC might lead to either LOC or LOL event.

25.2.2.6 Stop mode acknowledge error (SACKERR)

The Stop mode acknowledge error (SACKERR) reset is generated if the core attempts to enter Stop mode, but not all modules acknowledge stop mode within (2^16 +1 (corresponding to 0.5s)) cycles of the LPO128K_CLK.

A module might not acknowledge the entry to Stop mode if an error condition occurs. The error can be caused by a failure of an external clock input to a module.

The RCM_SRS[SACKERR] bit is written to 1 to indicate this reset source.

25.2.2.7 Software reset (SW)

The SYSRESETREQ bit in the NVIC application interrupt and reset control register can be written to 1 to force a software reset. (See Arm's NVIC documentation for the full description of the register fields, especially the VECTKEY field requirements.) Writing 1 to SYSRESETREQ generates a software reset request. This reset forces a system reset of all major components except for the debug module. A software reset causes the MCU to write 1 to RCM_SRS[SW] field.

25.2.2.8 Lockup reset (LOCKUP)

The LOCKUP gives immediate indication of seriously errant kernel software. This is the result of the core being locked because of an unrecoverable exception following the activation of the processor’s built in system state protection hardware.

The LOCKUP condition causes a system reset and also causes the MCU to write 1 to RCM_SRS[LOCKUP] field. See Arm Cortex-M4 Devices Generic User Guide (for S32K14x devices) and Arm Cortex-M0 Devices Generic User Guide (for S32K11x devices) for details on 'lockup' event.

25.2.2.9 MDM-AP system reset request

Writing 1 to the system reset request field in the MDM-AP control register initiates a system reset. This is the primary method for resets via the JTAG/SWD interface. The system reset is held until this field is written to 0.

Writing 1 to the core hold reset bit in the MDM-AP control register holds the core in reset as the rest of the chip comes out of system reset.

25.2.3 MCU Resets

A variety of resets are generated by the MCU to reset different modules.

25.2.3.1 POR Only

The POR Only reset asserts on the POR reset source only. It resets the PMC registers.

The POR Only reset also causes all other reset types to occur.

25.2.3.2 Chip POR

The Chip POR asserts on POR, LVR or LVD.

The Chip POR also causes the chip reset (including early chip reset) to occur.

25.2.3.3 Early chip reset

The early chip reset asserts on all reset sources. It resets only the flash memory module. It deasserts before flash memory initialization begins (earlier than the chip reset deassertion).

25.2.3.4 Chip reset

Chip reset asserts on all reset sources and only deasserts after flash memory initialization has completed and the RESET_B pin has also negated.

25.2.3.5 Core reset

The core comes out of reset after one cycle of chip reset. The core can be held in reset using the core hold reset bit in the MDM-AP control register as the rest of the chip comes out of system reset.

25.2.4 Reset pin

For all reset sources, the RESET_B pin is driven low by the MCU for at least 128 bus clock cycles and until flash memory initialization has completed.

After flash memory initialization has completed, the RESET_B pin is released and the internal chip reset deasserts. Keeping the RESET_B pin asserted externally delays the negation of the internal chip reset.

25.2.5 Debug resets

The following sections detail the debug resets.

25.2.5.1 JTAG reset

The JTAG module generate a system reset when certain IR codes are selected. This functional reset is asserted when EXTEST, HIGHZ and CLAMP instructions are active. The reset source from the JTAG module is released when any other IR code is selected. A JTAG reset causes the MCU to write 1 to RCM_SRS[JTAG] field.

25.2.5.2 Resetting the debug subsystem

Use the CDBGRSTREQ field within the SWJ-DP CTRL/STAT register to reset the debug modules. However the CDBGRSTREQ field does not reset all debug-related registers.

CDBGRSTREQ resets only the debug-related registers within the following modules:

- SWJ-DP
- AHB-AP
- TPIU
- MDM-AP (MDM control and status registers)

CDBGRSTREQ does not reset the debug-related registers within these modules:

- CM4 core (core debug registers: DHCSR, DCRSR, DCRDR, DEMCR)
- FPB
- DWT
- ITM
- NVIC
- Crossbar bus switch
- Private peripheral bus

25.3 Boot

This section describes the boot sequence, including sources and options.

25.3.1 Boot sources

This MCU supports cold booting from internal flash memory.

When the MCU boots from internal flash memory, the reset vectors are located at address 0x0 (initial SP_main) and 0x4 (initial PC).

The MCU also supports boot from RAM by relocating the exception vector table to RAM. This is implemented through a programmable Vector Table Offset Register (VTOR) in NVIC module.

The MCU supports serial code download via
- CAN
- LIN
- SPI etc.

The application developer can write their own flash memory based boot loaders or adapt the ones available at http://www.nxp.com. These boot loaders are then used to configure and use the communication protocol to perform mass serial download. For instance, the CAN module can be used to transfer data into its own message buffers, then the DMA module can be used to move the data into the main system RAM. Once the download is completed, the core can be used to transfer the new downloaded code into the flash memory area.

25.3.2 FOPT boot options

The FOPT register in the Flash Memory module allows the user to customize the operation of the MCU at boot time. The register contains read-only bits that are loaded from the NVM's option byte in the flash memory configuration field. The default setting for all values in the FTFC_FOPT register is logic 1 since it is copied from the option byte residing in flash memory, which has all bits as logic 1 in the flash memory erased state. To configure for alternate settings, program the appropriate bits in the NVM option byte. The new settings take effect on subsequent POR and any system reset. For more details on programming the option byte, see the flash memory chapter.

The MCU uses FOPT to configure the MCU at reset as shown in this table.

Table 25-2. FTFC_FOPT definition

Bit number	Field	Value	Definition
7-6	Reserved	Reserved for future expansion	
5	Reserved	Reserved	
4	Reserved	Reserved for future expansion	
3	RESET_PIN_CFG	Enables/disables control for the RESET pin.	
		0	RESET_B pin is disabled following a POR and cannot be enabled as reset function. When this option is selected, there could be a short period of contention during a POR ramp where the MCU drives the pin low prior to establishing the setting of this option and releasing the reset function on the pin. This bit is preserved through system resets and low-power modes. When RESET_B pin function is disabled, it cannot be used as a source for low-power mode wake-up.
		1	The port is configured with pullup enabled, passive filter enabled.
2	NMI_PIN_CFG	Enables/disables control for the NMI function.	
		0	NMI interrupts are always blocked. The associated pin continues to default to NMI_b pin controls with internal pullup enabled. When NMI_b pin function is disabled, it cannot be used as a source for low-power mode wake-up. If the NMI function is not required, either for an interrupt or wake up source, it is recommended that the NMI function be disabled by writing 0 to NMI_PIN_CFG.
		1	NMI_b pin/interrupts reset default to enabled.
1	Reserved	Reserved for future expansion.	
0	LPBOOT[1]	Controls the reset value of clock divider of IRC 48 Mhz to feed the Core and platform clocks. Larger divide value selections produce lower average power consumption during POR and reset sequencing and after reset exit.	
		0	Core and system clock divider (OUTDIV1) is 0x1 (divide by 2).
		1	Core and system clock divider (OUTDIV1) is 0x0 (divide by 1).

1. Not available in the S32K14xW devices

25.3.3 Boot sequence

At power up, the on-chip regulator holds the system in a POR state until the input supply is above the POR threshold. The system continues to be held in this static state until the internally regulated supplies have reached a safe operating voltage as determined by the LVD. The mode controller reset logic then controls this sequence to exit reset:

1. A system reset is held on internal logic, the RESET_B pin is driven out low, and the SCG is enabled in its default clocking mode.
2. Required clocks are enabled (core clock, system clock, flash clock, and any bus clocks that do not have clock gate control reset to disabled).

3. The system reset on internal logic continues to be held, but the flash controller is released from reset and begins initialization operation while the reset control logic continues to drive the RESET_B pin low.
4. Early in reset sequencing, the NVM option byte is read and stored to the flash memory module's FOPT register. If the LPBOOT is programmed for an alternate clock divider reset value, the system/core clock is switched to a slower clock speed.
5. When flash memory initialization completes, the RESET_B pin is released. If RESET_B continues to be asserted (an indication of a slow rise time on the RESET_B pin or external drive in low), the system continues to be held in reset. Once the RESET_B pin is detected high, the core clock is enabled and the system is released from reset.
6. When the system exits reset, the processor sets up the stack, program counter (PC), and link register (LR). The processor reads the start SP (SP_main) from vector-table offset 0. The core reads the start PC from vector-table offset 4. LR is written to 0xFFFF_FFFF.
7. If FlexNVM is enabled, the flash memory controller continues to restore the FlexNVM data. This data is not available immediately out of reset and the system should not access this data until the flash controller completes this initialization step as indicated by the EEERDY flag.

Subsequent system resets follow this same reset flow.

Chapter 26
Reset Control Module (RCM)

26.1 Chip-specific RCM information

High voltage detect (HVD) reset is not supported on this device. References to HVD reset or HVD trip must be ignored. The SACKERR timing is 0.5 s.

On this device, LVD (Low voltage detect) and LVR (Low voltage reset) events trigger LVD reset and hence Chip POR. Chip POR is triggered by POR, LVR or LVD events. The PMC_LVDSC1[LVDRE] is used to gate LVD only, and has no effect on LVR generation. With PMC_SC1[LVDRE] disabled, LVRs can still generate reset to system. See Low Voltage Detect (LVD) System.

Wait mode is not supported on this device. See Module operation in available power modes for details on available power modes.

LPO128K_CLK is used as low power clock for reset pin filter.

See ARM Cortex-M4 Devices Generic User Guide (for S32K14x and S32K14xW devices) and ARM Cortex-M0 Devices Generic User Guide (for S32K11x devices) for details on 'lockup' event.

26.1.1 RCM register information

Following bit fields details CMU reset source and are available in S32K11x variants only:

- RCM_PARAM[ECMU_LOC],
- RCM_SRS[CMU_LOC],
- RCM_SSRS[SCMU_LOC], and
- RCM_SRIE[CMU_LOC].

Following bit fields details reset caused by a loss of lock in the SCG PLL/FLL and are available in S32K14x and S32K14xW variants only:

- RCM_PARAM[ELOL],

- RCM_SRS[LOL], and
- RCM_SSRS[SLOL].

NOTE

The read of the RCM registers is affected during EEERAM commands. For more information see note 'During execution of the SetRAM 0xFF' in Chip-specific FTFC information.

26.2 Reset pin filter operation in STOP1/2 modes

The reset pins filtering in STOP1/2 mode is as follows:

- STOP1: Both RUN mode and STOP mode filters can operate. The filter with lower pulse duration dominates.
- STOP2: RUN mode filter operates.

26.3 Introduction

Information found here describes the registers of the Reset Control Module (RCM). The RCM implements many of the reset functions for the chip. See the chip's reset chapter for more information.

See AN4503: Power Management for Kinetis and ColdFire+ MCUs for further details on using the RCM.

26.4 Reset memory map and register descriptions

The RCM Memory Map/Register Definition can be found here.

The Reset Control Module (RCM) registers provide reset status information and reset filter control.

NOTE

The RCM registers can be written only in supervisor mode. Write accesses in user mode are blocked and will result in a bus error.

RCM memory map

Address offset (hex)	Register name	Width (in bits)	Access	Reset value	Section/ page
0	Version ID Register (RCM_VERID)	32	R	0300_0003h	26.4.1/541
4	Parameter Register (RCM_PARAM)	32	R	See section	26.4.2/542
8	System Reset Status Register (RCM_SRS)	32	R	0000_0082h	26.4.3/544
C	Reset Pin Control register (RCM_RPC)	32	R/W	0000_0000h	26.4.4/547
18	Sticky System Reset Status Register (RCM_SSRS)	32	R/W	0000_0082h	26.4.5/549
1C	System Reset Interrupt Enable Register (RCM_SRIE)	32	R/W	0000_0000h	26.4.6/551

26.4.1 Version ID Register (RCM_VERID)

Address: 0h base + 0h offset = 0h

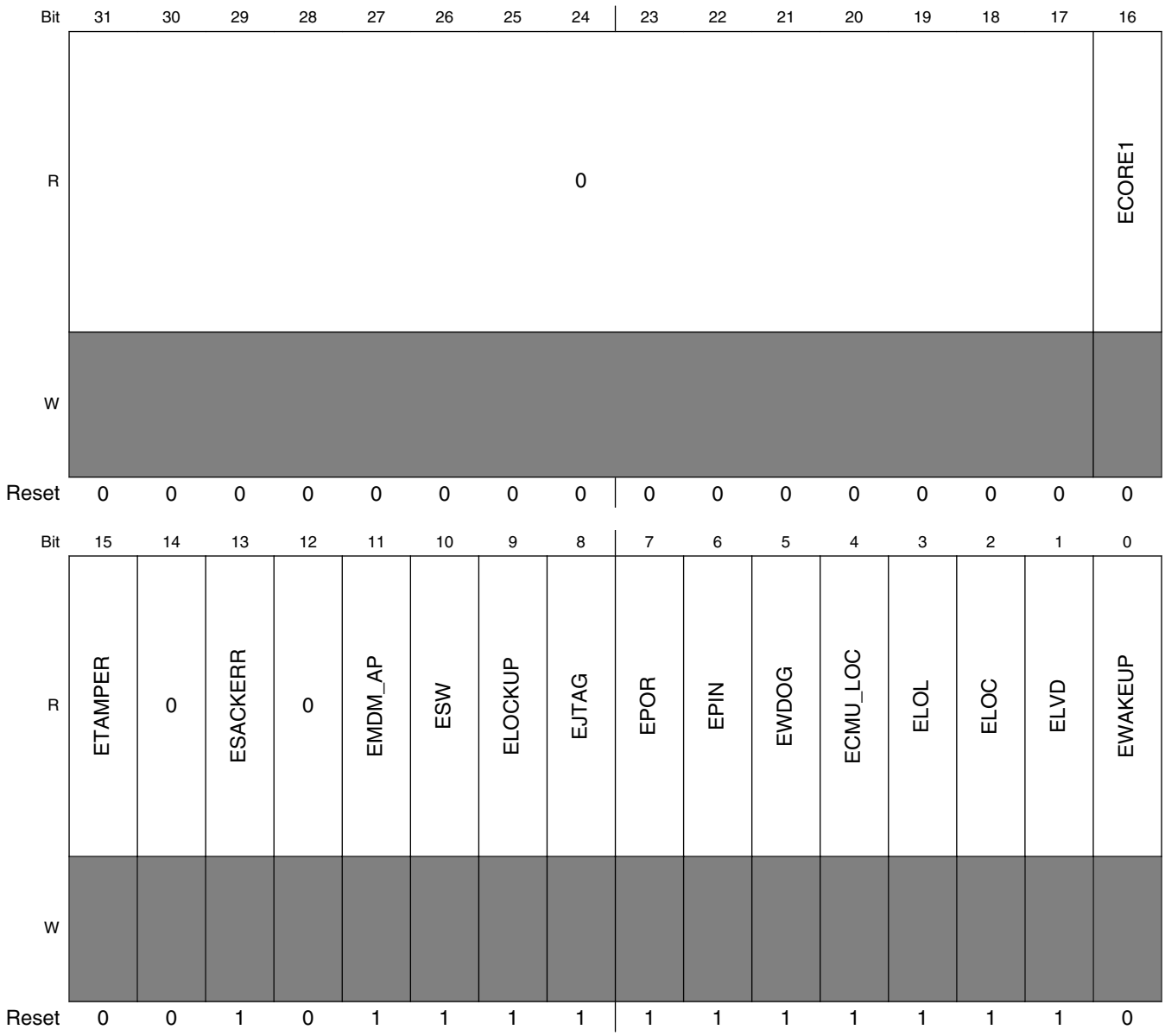
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	MAJOR								MINOR								FEATURE															
W																																
Reset	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

RCM_VERID field descriptions

Field	Description
31–24 MAJOR	Major Version Number This read only field returns the major version number for the specification.
23–16 MINOR	Minor Version Number This read only field returns the minor version number for the specification.
FEATURE	Feature Specification Number This read only field returns the feature set number. 0x0003 Standard feature set.

26.4.2 Parameter Register (RCM_PARAM)

Address: 0h base + 4h offset = 4h



RCM_PARAM field descriptions

Field	Description
31–17 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
16 ECORE1	Existence of SRS[CORE1] status indication feature

Table continues on the next page...

RCM_PARAM field descriptions (continued)

Field	Description
	This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
15 ETAMPER	Existence of SRS[TAMPER] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
14 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
13 ESACKERR	Existence of SRS[SACKERR] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
12 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
11 EMDM_AP	Existence of SRS[MDM_AP] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
10 ESW	Existence of SRS[SW] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
9 ELOCKUP	Existence of SRS[LOCKUP] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
8 EJTAG	Existence of SRS[JTAG] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
7 EPOR	Existence of SRS[POR] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.

Table continues on the next page...

RCM_PARAM field descriptions (continued)

Field	Description
6 EPIN	Existence of SRS[PIN] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
5 EWDOG	Existence of SRS[WDOG] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
4 ECMU_LOC	Existence of SRS[CMU_LOC] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
3 ELOL	Existence of SRS[LOL] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
2 ELOC	Existence of SRS[LOC] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
1 ELVD	Existence of SRS[LVD] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.
0 EWAKEUP	Existence of SRS[WAKEUP] status indication feature This static bit states whether or not the feature is available on the device. 0 The feature is not available. 1 The feature is available.

26.4.3 System Reset Status Register (RCM_SRS)

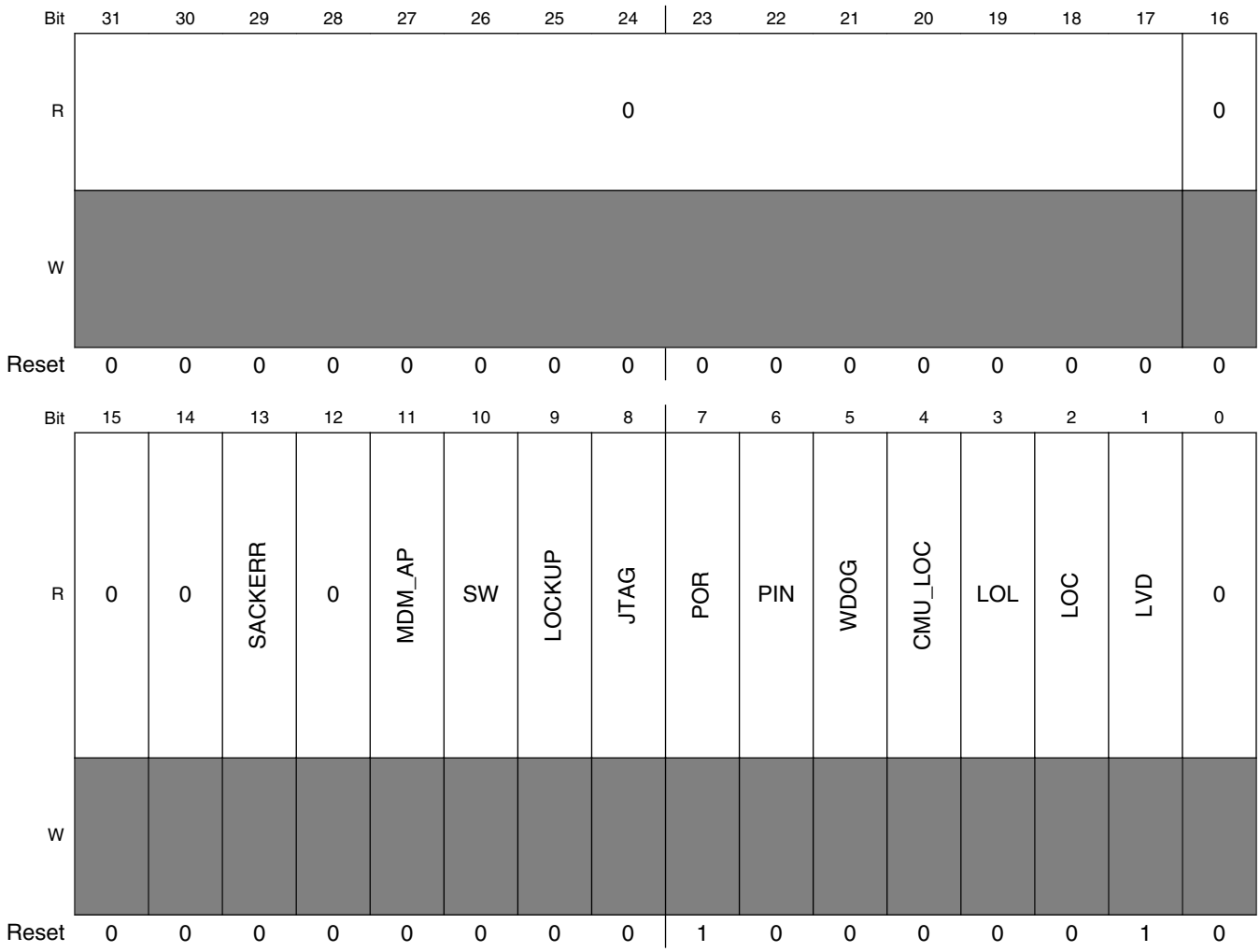
This register includes read-only status flags to indicate the source of the most recent reset. Note that multiple flags can be set if multiple reset events occur at the same time. The reset state of these bits depends on what caused the MCU to reset.

NOTE

The reset value of this register depends on the reset source:

- POR (including LVD) — 0x82
- LVD (without POR) — 0x02
- Other reset — a bit is set if its corresponding reset source caused the reset

Address: 0h base + 8h offset = 8h



RCM_SRS field descriptions

Field	Description
31–17 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
15 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

Table continues on the next page...

RCM_SRS field descriptions (continued)

Field	Description
14 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
13 SACKERR	Stop Acknowledge Error Indicates that after an attempt to enter Stop mode, a reset has been caused by a failure of one or more peripherals to acknowledge within approximately one second to enter stop mode. 0 Reset not caused by peripheral failure to acknowledge attempt to enter stop mode 1 Reset caused by peripheral failure to acknowledge attempt to enter stop mode
12 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
11 MDM_AP	MDM-AP System Reset Request Indicates a reset has been caused by the host debugger system setting of the System Reset Request bit in the MDM-AP Control Register. 0 Reset was not caused by host debugger system setting of the System Reset Request bit 1 Reset was caused by host debugger system setting of the System Reset Request bit
10 SW	Software Indicates a reset has been caused by software setting of SYSRESETREQ bit in Application Interrupt and Reset Control Register in the Arm core. 0 Reset not caused by software setting of SYSRESETREQ bit 1 Reset caused by software setting of SYSRESETREQ bit
9 LOCKUP	Core Lockup Indicates a reset has been caused by the Arm core indication of a LOCKUP event. 0 Reset not caused by core LOCKUP event 1 Reset caused by core LOCKUP event
8 JTAG	JTAG generated reset Indicates a reset has been caused by the JTAG selection of certain IR codes: EXTEST, HIGHZ, and CLAMP. 0 Reset not caused by JTAG 1 Reset caused by JTAG
7 POR	Power-On Reset Indicates a reset has been caused by the power-on detection logic. Because the internal supply voltage was ramping up at the time, the low-voltage reset (LVD) status bit is also set to indicate that the reset occurred while the internal supply was below the LVD threshold. 0 Reset not caused by POR 1 Reset caused by POR
6 PIN	External Reset Pin Indicates a reset has been caused by an active-low level on the external $\overline{\text{RESET}}$ (RESET_b) pin. 0 Reset not caused by external reset pin 1 Reset caused by external reset pin

Table continues on the next page...

RCM_SRS field descriptions (continued)

Field	Description
5 WDOG	Watchdog Indicates a reset has been caused by the watchdog timer timing out. This reset source can be blocked by disabling the watchdog. 0 Reset not caused by watchdog timeout 1 Reset caused by watchdog timeout
4 CMU_LOC	CMU Loss-of-Clock Reset Indicates a reset has been caused by the CMU loss-of-clock circuit. CMU_FC_GCR[FCE] must be set to enable the frequency check circuit. See the CMU chapter for details. 0 Reset not caused by the CMU loss-of-clock circuit. 1 Reset caused by the CMU loss-of-clock circuit.
3 LOL	Loss-of-Lock Reset Indicates a reset has been caused by a loss of lock in the SCG PLL/FLL. 0 Reset not caused by a loss of lock in the PLL/FLL 1 Reset caused by a loss of lock in the PLL/FLL
2 LOC	Loss-of-Clock Reset Indicates a reset has been caused by a loss of external clock. The SCG SOSC clock monitor must be enabled for a loss of clock to be detected. Refer to the detailed SCG description for information on enabling the clock monitor. 0 Reset not caused by a loss of external clock. 1 Reset caused by a loss of external clock.
1 LVD	Low-Voltage Detect Reset or High-Voltage Detect Reset If PMC_LVDSC1[LVDRE] is set and the supply drops below the LVD trip voltage, an LVD reset occurs. If PMC_HVDSC1[HVDRE] is set and the supply rises above the HVD trip voltage, an HVD reset occurs. This field is also set by POR. 0 Reset not caused by LVD trip, HVD trip or POR 1 Reset caused by LVD trip, HVD trip or POR
0 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

26.4.4 Reset Pin Control register (RCM_RPC)

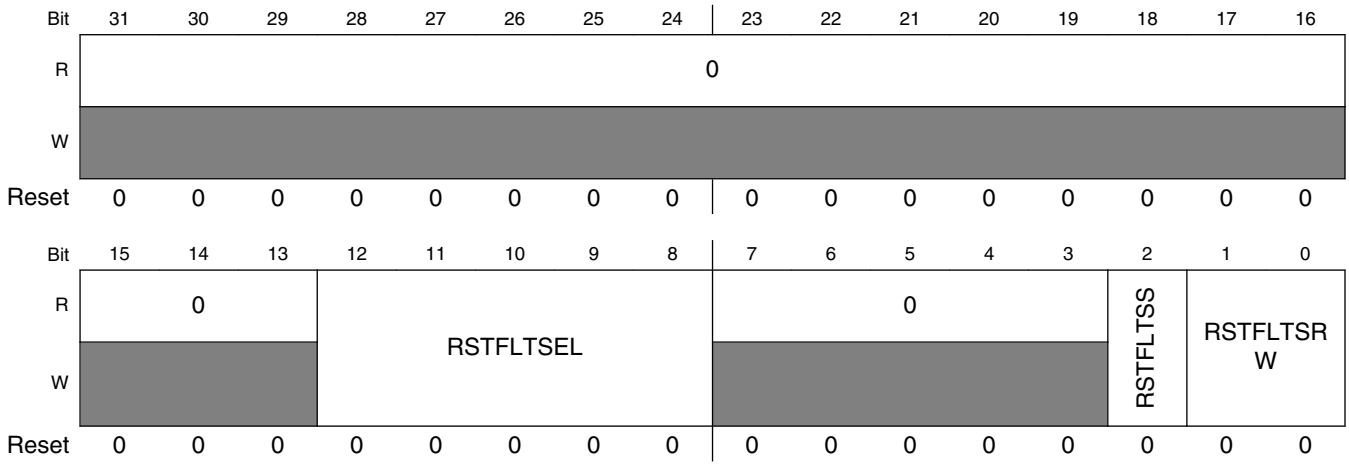
NOTE

This register is reset on Chip POR only, it is unaffected by other reset types.

NOTE

The bus clock filter is reset when disabled or when entering stop mode. The LPO filter is reset when disabled.

Address: 0h base + Ch offset = Ch



RCM_RPC field descriptions

Field	Description
31–13 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
12–8 RSTFLTSEL	Reset Pin Filter Bus Clock Select Selects the reset pin bus clock filter width: • Transition lengths less than RSTFLTSEL cycles are filtered. • Transition lengths between RSTFLTSEL and (RSTFLTSEL+1) cycles (inclusive) may be filtered. • Transition lengths greater than (RSTFLTSEL+1) cycles are not filtered.
7–3 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
2 RSTFLTSS	Reset Pin Filter Select in Stop Mode Selects how the reset pin filter is enabled in any stop mode. 0 All filtering disabled 1 LPO clock filter enabled
RSTFLTSRW	Reset Pin Filter Select in Run and Wait Modes Selects how the reset pin filter is enabled in run and wait modes. 00 All filtering disabled 01 Bus clock filter enabled for normal operation 10 LPO clock filter enabled for normal operation 11 Reserved

26.4.5 Sticky System Reset Status Register (RCM_SSRS)

This register includes status flags to indicate all reset sources since the last POR or LVD that have not been cleared by software. Software can clear the status flags by writing a logic one to a flag.

Address: 0h base + 18h offset = 18h

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															0
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0	0	SSACKERR	0	SMDM_AP	SSW	SLOCKUP	SJTAG	SPOR	SPIN	SWDOG	SCMU_LOC	SLOL	SLOC	SLVD	0
W			w1c		w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	w1c	
Reset	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0

RCM_SSRS field descriptions

Field	Description
31–17 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
15 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
14 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
13 SSACKERR	Sticky Stop Acknowledge Error

Table continues on the next page...

RCM_SSRS field descriptions (continued)

Field	Description
	Indicates that after an attempt to enter Stop mode, a reset has been caused by a failure of one or more peripherals to acknowledge within approximately one second to enter stop mode. 0 Reset not caused by peripheral failure to acknowledge attempt to enter stop mode 1 Reset caused by peripheral failure to acknowledge attempt to enter stop mode
12 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
11 SMDM_AP	Sticky MDM-AP System Reset Request Indicates a reset has been caused by the host debugger system setting of the System Reset Request bit in the MDM-AP Control Register. 0 Reset was not caused by host debugger system setting of the System Reset Request bit 1 Reset was caused by host debugger system setting of the System Reset Request bit
10 SSW	Sticky Software Indicates a reset has been caused by software setting of SYSRESETREQ bit in Application Interrupt and Reset Control Register in the Arm core. 0 Reset not caused by software setting of SYSRESETREQ bit 1 Reset caused by software setting of SYSRESETREQ bit
9 SLOCKUP	Sticky Core Lockup Indicates a reset has been caused by the Arm core indication of a LOCKUP event. 0 Reset not caused by core LOCKUP event 1 Reset caused by core LOCKUP event
8 SJTAG	Sticky JTAG generated reset Indicates a reset has been caused by the JTAG selection of certain IR codes: EXTEST, HIGHZ, and CLAMP. 0 Reset not caused by JTAG 1 Reset caused by JTAG
7 SPOR	Sticky Power-On Reset Indicates a reset has been caused by the power-on detection logic. Because the internal supply voltage was ramping up at the time, the low-voltage reset (LVD) status bit is also set to indicate that the reset occurred while the internal supply was below the LVD threshold. 0 Reset not caused by POR 1 Reset caused by POR
6 SPIN	Sticky External Reset Pin Indicates a reset has been caused by an active-low level on the external $\overline{\text{RESET}}$ (RESET_b) pin. 0 Reset not caused by external reset pin 1 Reset caused by external reset pin
5 SWDOG	Sticky Watchdog Indicates a reset has been caused by the watchdog timer timing out. This reset source can be blocked by disabling the watchdog.

Table continues on the next page...

RCM_SSRS field descriptions (continued)

Field	Description
	0 Reset not caused by watchdog timeout 1 Reset caused by watchdog timeout
4 SCMU_LOC	Sticky CMU Loss-of-Clock Reset Indicates a reset has been caused by the CMU loss-of-clock circuit. CMU_FC_GCR[FCE] must be set to enable the frequency check circuit. See the CMU chapter for details. 0 Reset not caused by the CMU loss-of-clock circuit. 1 Reset caused by the CMU loss-of-clock circuit.
3 SLOL	Sticky Loss-of-Lock Reset Indicates a reset has been caused by a loss of lock in the SCG PLL/FLL. See the SCG description for information on the loss-of-lock event. 0 Reset not caused by a loss of lock in the PLL/FLL 1 Reset caused by a loss of lock in the PLL/FLL
2 SLOC	Sticky Loss-of-Clock Reset Indicates a reset has been caused by a loss of external clock. The SCG SOSC clock monitor must be enabled for a loss of clock to be detected. Refer to the detailed SCG description for information on enabling the clock monitor. 0 Reset not caused by a loss of external clock. 1 Reset caused by a loss of external clock.
1 SLVD	Sticky Low-Voltage Detect Reset If PMC_LVDSC1[LVDRE] is set and the supply drops below the LVD trip voltage, an LVD reset occurs. This field is also set by POR. 0 Reset not caused by LVD trip or POR 1 Reset caused by LVD trip or POR
0 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

26.4.6 System Reset Interrupt Enable Register (RCM_SRIE)

This register provides the option to delay the assertion of a system reset for a period of time (DELAY field) while an interrupt is generated. When an interrupt for a reset source is enabled, software has time to perform a graceful shutdown. A Chip POR source cannot be delayed by this feature. The SRS updates only after the system reset occurs.

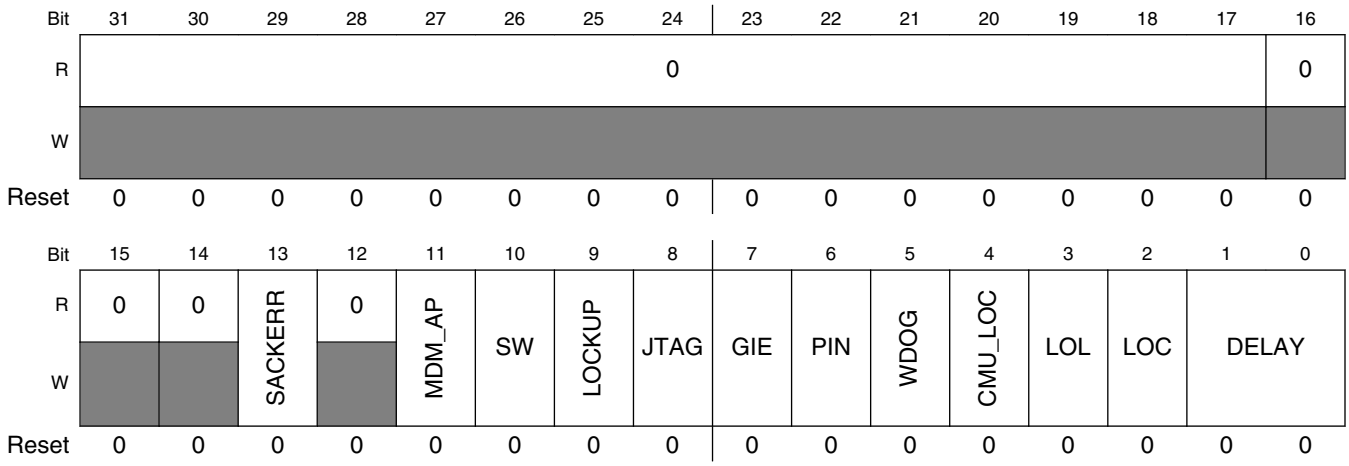
NOTE

The reset delay feature requires the LPO clock to remain active.

NOTE

This register is reset on Chip POR only, it is unaffected by other reset types.

Address: 0h base + 1Ch offset = 1Ch



RCM_SRIE field descriptions

Field	Description
31–17 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
15 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
14 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
13 SACKERR	Stop Acknowledge Error Interrupt 0 Interrupt disabled. 1 Interrupt enabled.
12 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
11 MDM_AP	MDM-AP System Reset Request 0 Interrupt disabled. 1 Interrupt enabled.
10 SW	Software Interrupt 0 Interrupt disabled. 1 Interrupt enabled.
9 LOCKUP	Core Lockup Interrupt **NOTE:** The LOCKUP bit is useful only in devices with more than one core processor. 0 Interrupt disabled. 1 Interrupt enabled.
8 JTAG	JTAG generated reset 0 Interrupt disabled. 1 Interrupt enabled.

Table continues on the next page...

RCM_SRIE field descriptions (continued)

Field	Description
7 GIE	Global Interrupt Enable 0 All interrupt sources disabled. 1 All interrupt sources enabled. Note that the individual interrupt-enable bits still need to be set to generate interrupts.
6 PIN	External Reset Pin Interrupt 0 Reset not caused by external reset pin 1 Reset caused by external reset pin
5 WDOG	Watchdog Interrupt 0 Interrupt disabled. 1 Interrupt enabled.
4 CMU_LOC	CMU Loss-of-Clock Interrupt 0 Interrupt disabled. 1 Interrupt enabled.
3 LOL	Loss-of-Lock Interrupt 0 Interrupt disabled. 1 Interrupt enabled.
2 LOC	Loss-of-Clock Interrupt 0 Interrupt disabled. 1 Interrupt enabled.
DELAY	Reset Delay Time Configures the maximum reset delay time from when the interrupt is asserted and the system reset occurs. 00 10 LPO cycles 01 34 LPO cycles 10 130 LPO cycles 11 514 LPO cycles